

Solids



1 Name the following solids:

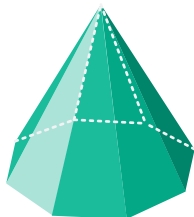
a



b



c



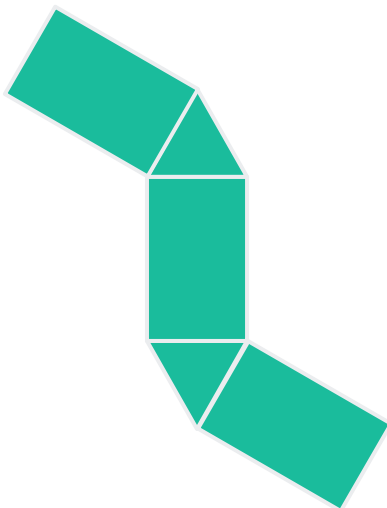
d



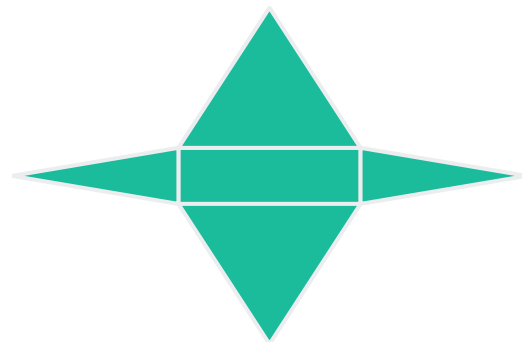
Nets

2 Name the solid formed from each of the following nets:

a



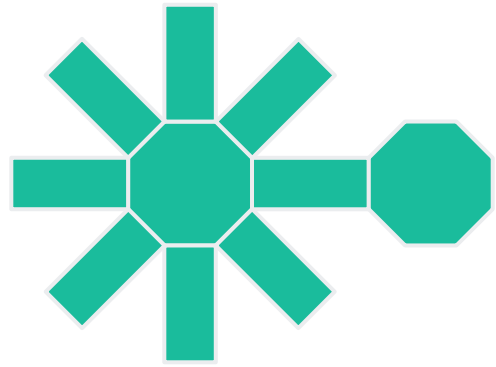
b



c



d



3 Draw a net for the following solids:

a



b



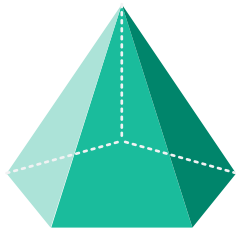
c



d



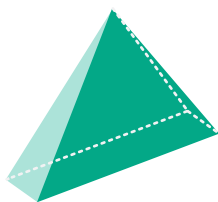
e



f



g

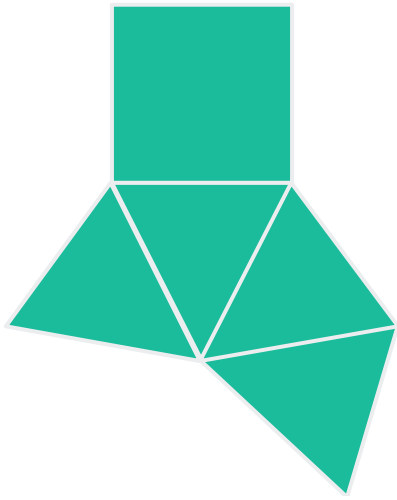


h



4 Name the 3D shapes formed by the following nets:

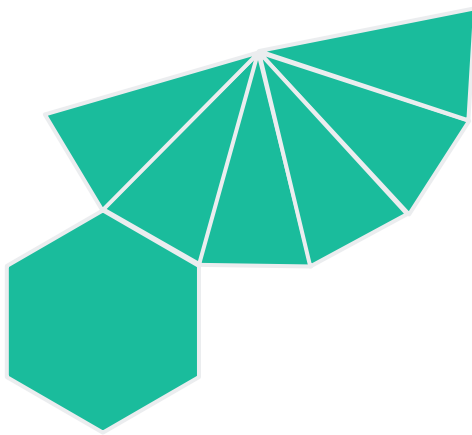
a



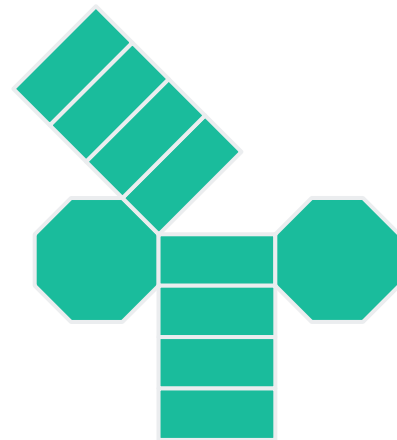
b



c



d

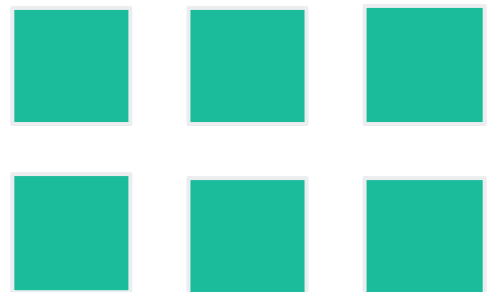


5 State the solid that can be formed from the given sets of 2D shapes:

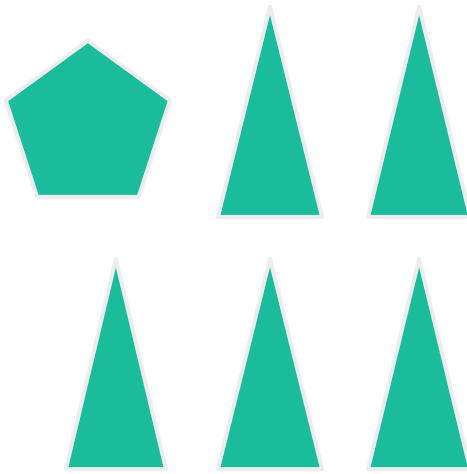
a



b



c



Front, side, and plan view

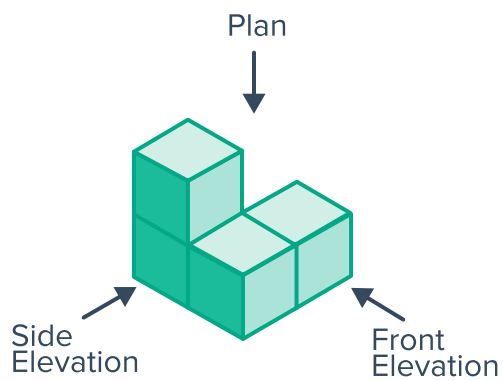
6 For each of the following solids formed from cubes:

i Draw the diagram for the side view.

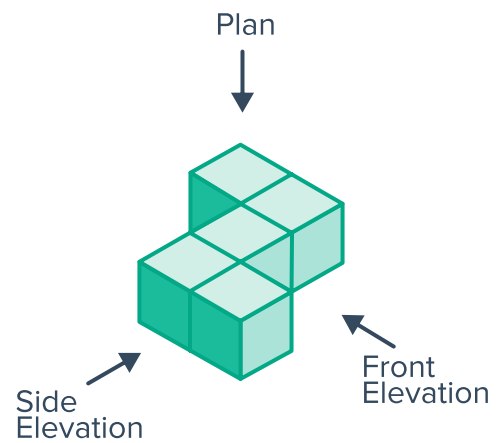
ii Draw the diagram for the front view.

iii Draw the diagram for the plan view.

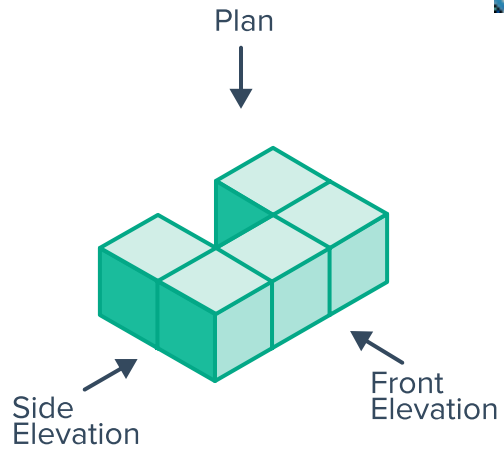
a



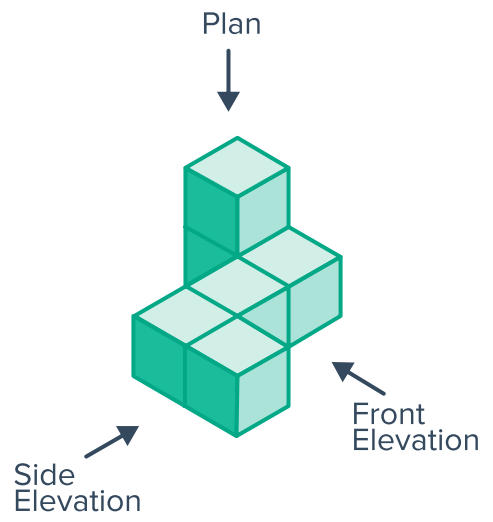
b



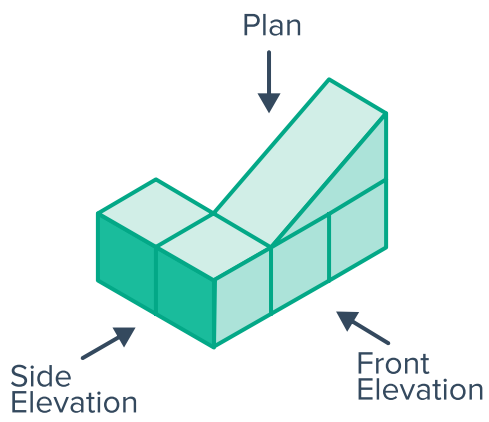
c



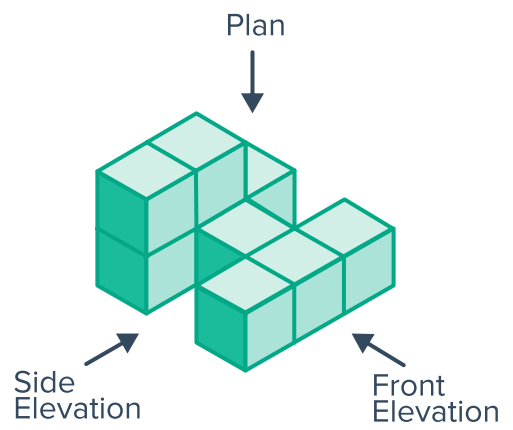
d



e



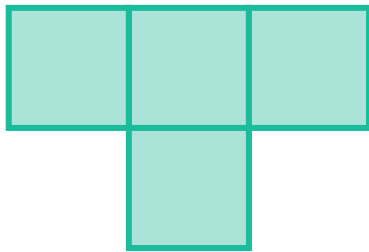
f



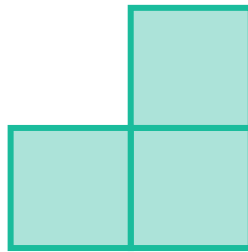
7 Select the solid that matches the given elevations:



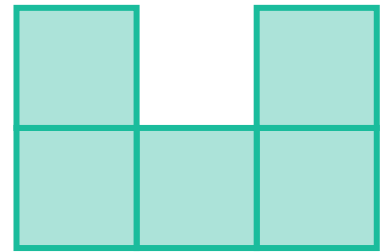
a



Plan

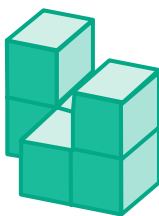


Front Elevation

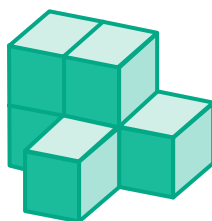


Side Elevation

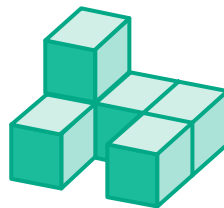
A



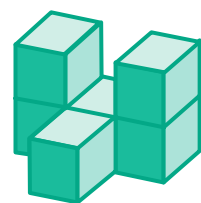
B



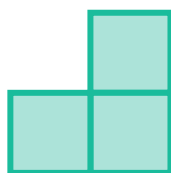
C



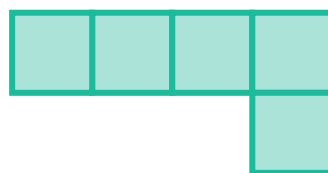
D



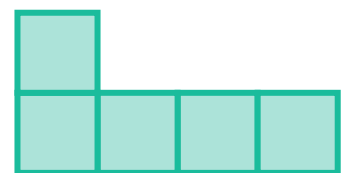
b



Front Elevation

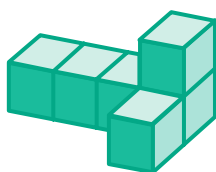


Plan



Side Elevation

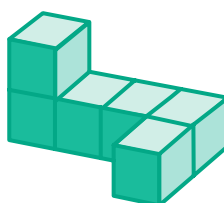
A



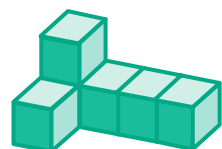
B



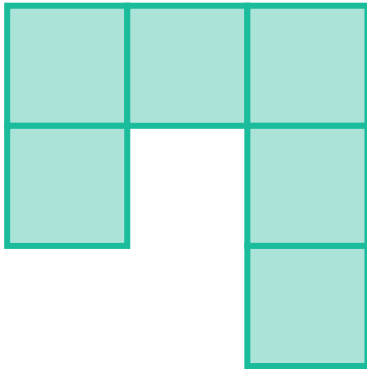
C



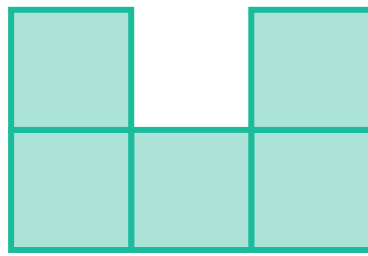
D



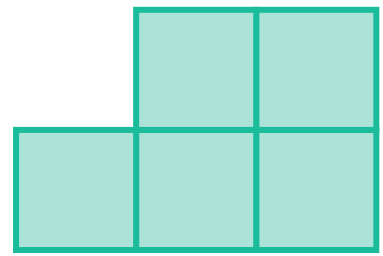
c



Plan

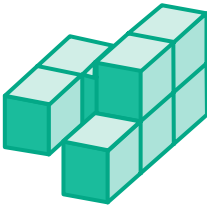


Side Elevation

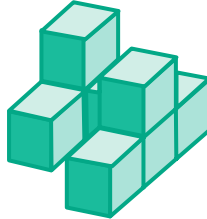


Front Elevation

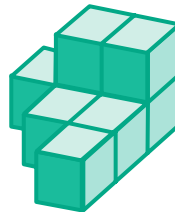
A



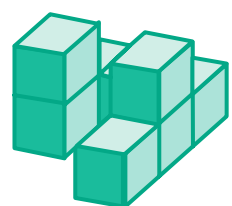
B



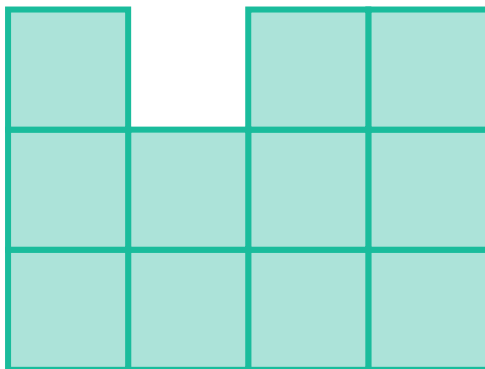
C



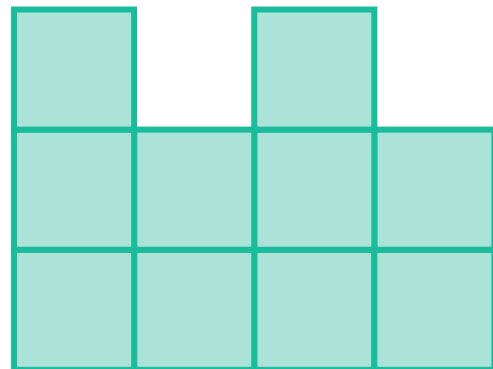
D



d

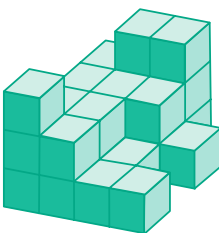


Front Elevation

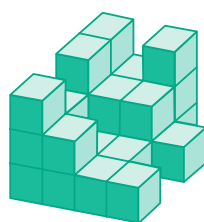


Side Elevation

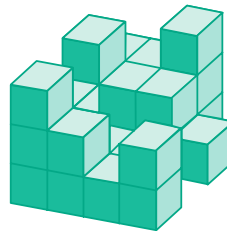
A



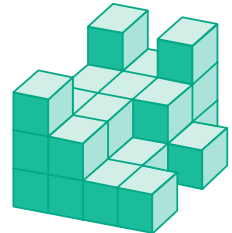
B



C



D



8 Name the following solids:



a



b



c



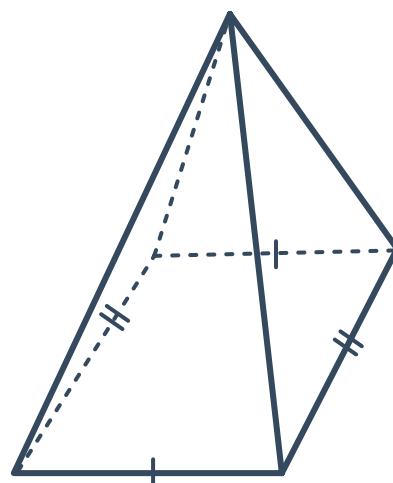
9 For each of the following solids:

- i Name the solid.
- ii State whether the solid has a uniform cross-section.
- iii State the order of rotational symmetry about a vertical axis down the center of the solid.
- iii State whether the solid has plane symmetry.

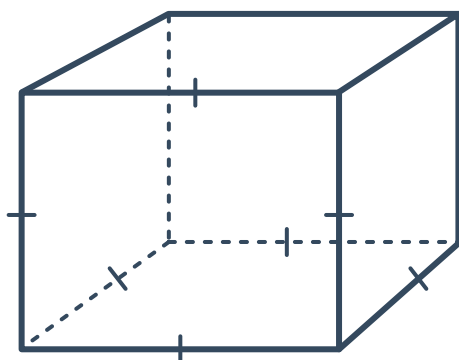
a



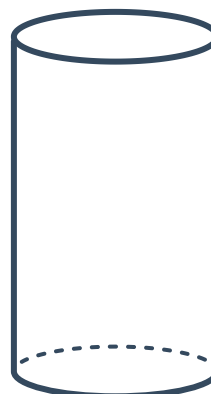
b



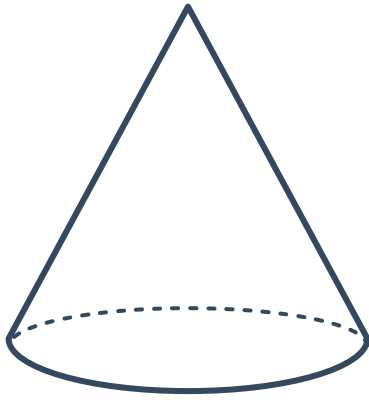
c



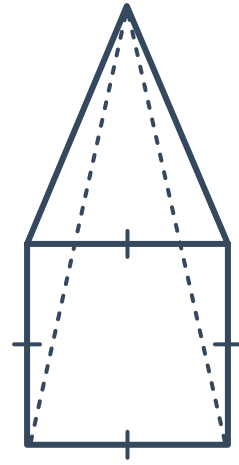
d



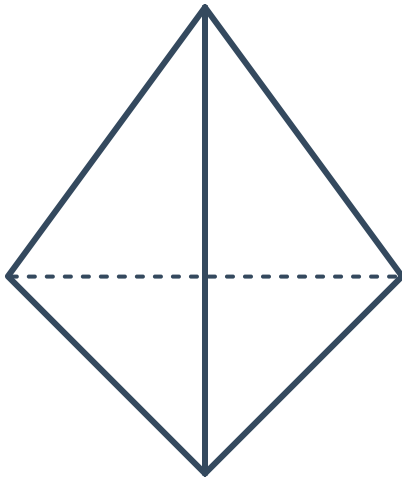
e



f



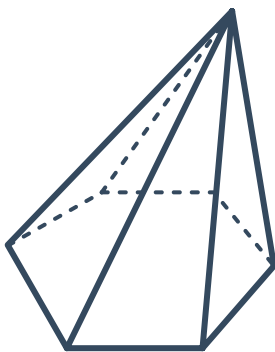
g



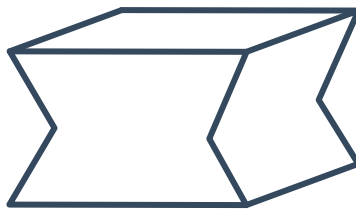
h



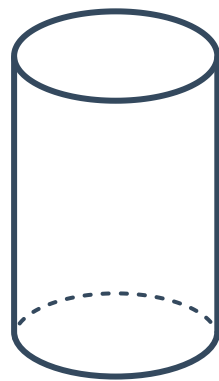
10 Explain why shapes A and B are polyhedra while shape C is not.



A



B

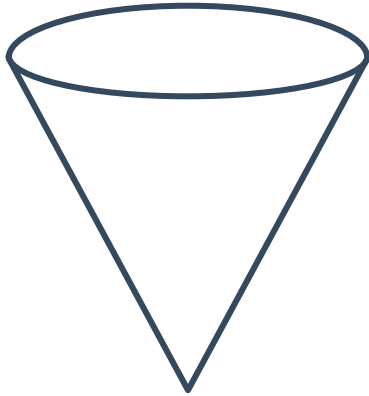


C

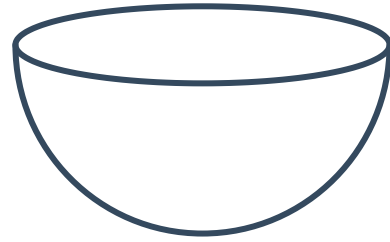
11 Determine whether the following solids are polyhedrons:



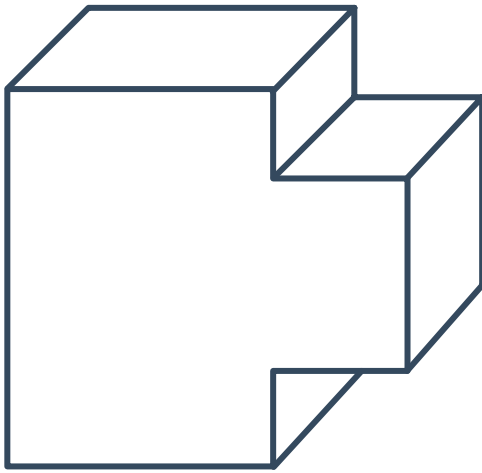
a



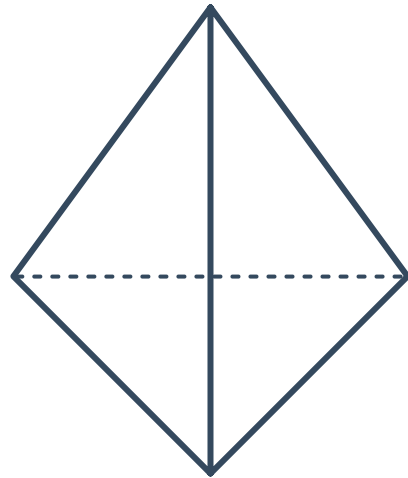
b



c



d



12 State whether the following solids have faces, vertices **and** edges:

a Sphere

b Cone

c Tetrahedron

d Cylinder

13 Consider a rectangular prism.

a How many faces meet at any one vertex?

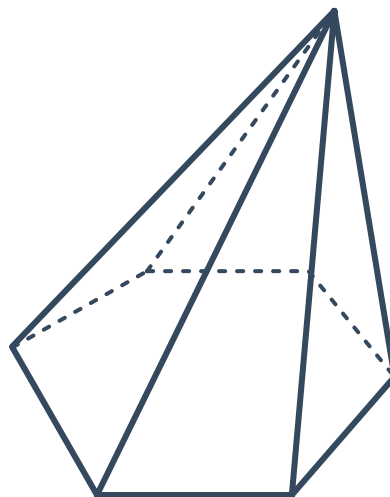
b How many faces meet at any one edge?

c How many edges meet at any one vertex?

14 Consider the following hexagonal pyramid



- a How many faces does it have?
- b How many vertices does it have?
- c How many edges does it have?

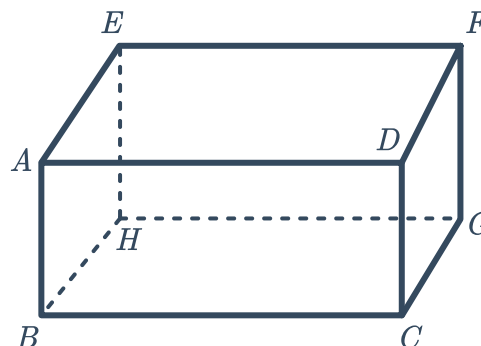


15 Complete the table with the number of features for each solid:

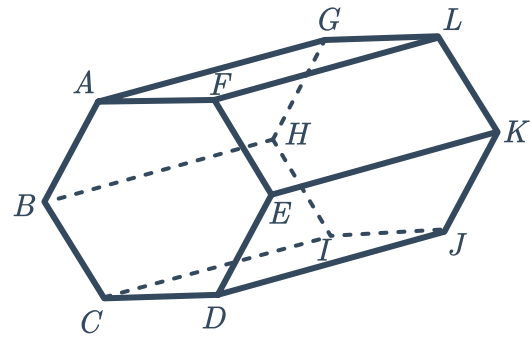
3D shape	Number of faces	Number of vertices	Number of edges
Hexagonal prism			
Triangular pyramid			
Rectangular prism			

16 Consider given the rectangular prism:

- a Name two parallel edges.
- b Name two edges that are both intersecting and perpendicular.
- c Name all of the edges that are parallel to BH .
- d Name all of the edges that are perpendicular to BH .
- e Does the prism have plane symmetry about the plane $AFGB$?



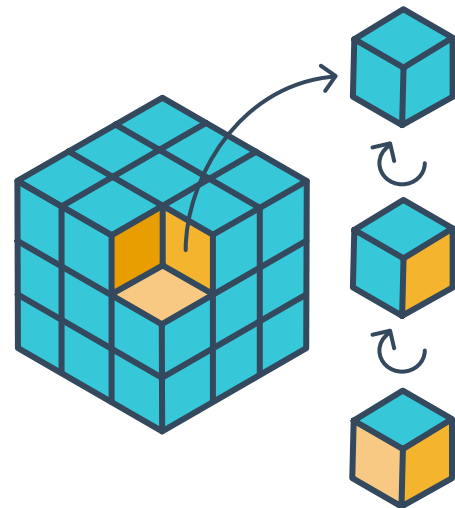
17 Consider the given hexagonal prism:



- a Name the face that is parallel to $FEKL$.
- b How many faces are perpendicular to $ABCDEF$?
- c Name two parallel edges.
- d Name two perpendicular edges.
- e State whether the following are planes of symmetry for the prism:
 - i $BEKH$
 - ii $ADJG$
 - iii $BDJH$
 - iv $CFLI$

18 A number of $1\text{ cm} \times 1\text{ cm} \times 1\text{ cm}$ cubes are glued together to form a $3\text{ cm} \times 3\text{ cm} \times 3\text{ cm}$ cube as shown:

The outside faces of the large cube are painted blue and the small cubes are then pulled apart.



- a How many of the small cubes will have at least one face painted blue?
- b How many of the small cubes will have at least two faces painted blue?
- c How many of the small cubes will have three faces painted blue?

19 Euler related the number of faces, edges and vertices of any polyhedron in one formula.

a Complete:

Solid	Number of Faces (F)	Number of Vertices (V)	Number of Edges (E)	$F + V$
Rectangular Prism		8		
Triangular Prism		6		
Hexagonal Prism		12		
Triangular Pyramid		4		
Pentagonal Pyramid		6		

b Determine whether each of the following rules correctly states the relationship between the number of faces, vertices and edges of a polyhedron:

i $F + V - E = 2$

ii $F + V + E = 2$

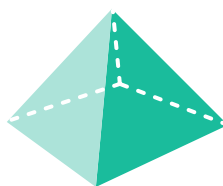
iii $E + V - F = 2$

iv $F + E - V = 2$

20 Select the right name for each solid from the list below:

- Square pyramid
- Rectangular pyramid
- Triangular prism
- Rectangular prism
- Octagonal pyramid
- Octagonal prism

a



b



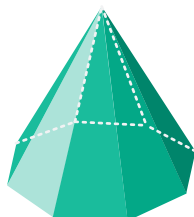
c



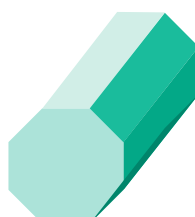
d



e



f



21 How many planes of symmetry do the following solids have?

a Cube

b Sphere



c Cone

d Square prism